Rebuilding a VO_2max mask for breathing measurements over time

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1. Abstract:

Reliable measurements of maximal oxygen uptake, VO_2max, are typically performed in laboratories using specialised equipment. This makes it difficult for people without access to these facilities to monitor their oxygen uptake and receive real-time feedback during exercise. The aim of this project is therefore to further develop a portable respiratory mask that could eventually enable the measurement and calculation of VO_2max and other respiratory parameters outside the laboratory. The project builds on an existing prototype consisting of a modified respiratory mask, a Venturi tube, and an electronics unit with an ESP32 microcontroller. The system's sensors measure oxygen and carbon dioxide concentrations, differential pressure, temperature, and relative humidity. These data will form the basis for calculating parameters such as airflow and oxygen uptake. The work primarily focuses on connecting and configuring the sensors and improving measurement stability. Initial tests show that the prototype can collect and record sensor data. However, measurement accuracy has not yet been established, and further calibration and validation are required before the system's ability to measure oxygen uptake can be assessed. Future work includes testing at rest and during physical activity and, if time permits, modifying the electronics unit to reduce its size and weight. The project is expected to provide a foundation for the continued development of a more accessible and practical system for respiratory monitoring during exercise.

2. Introduction

VO_2max represents the maximum rate at which the body can use oxygen during exercise and is a common indicator of aerobic fitness. It is typically measured using a metabolic cart connected to a respiratory mask that analyses oxygen consumption and carbon dioxide (CO_2) production. The main problem addressed in this project is how to rebuild and improve an existing VO_2max mask while ensuring that its sensors are correctly connected and configured to obtain reliable measurements [1].

Several methods are available for measuring respiratory gas exchange during exercise. Commercial metabolic carts combine gas analysers, airflow measurement, and data acquisition in one system [2]. Alternatively, modular systems use separate sensors to measure respiratory flow and gas concentrations [3].

The contributions of this project are the reconstruction and evaluation of an existing VO_2max mask prototype, focusing on sensor integration, system configuration, and measurement performance. The respiratory and CO_2 sensors were integrated into the system and configured for data collection. The results, due to various issues regarding both hardware and software, were inconclusive. The results provide a partial assessment of the prototype's functionality and identify areas for further improvement for potential use by Sports Technology at KTH.

3. Design

The mask-prototype consists of three modules: the (modified) particulate respirator mask, the venturi-tube and the electronic unit (which contains the battery, microcontroller, PCB and sensors as well as wires connecting the different components).

The particulate respirator mask is connected to the venturi-tube, allowing the user to breathe into it, and is worn comfortably by using its elastic bands to situate it over the lower half of one's face.

The venturi-tube is a type of tube with two different types of sections, the normal-width section and the narrowed section. The tube is constructed in this manner to be able to calculate the flow rate of the liquid (i.e. the gas) flowing through it. This works because of the phenomena known as the venturi-effect [4]: when a liquid passes through a narrowed passage the drop in pressure results in an increased velocity. By forcing the fluid to flow through a constricted section of the tube and measuring the pressure difference the flow rate of that fluid can be calculated.

The electronic unit contains the battery which supplies the microcontroller (which in turn supplies the sensors). The battery in question is a 3.7 V 1100 Mah LiPo battery. The microcontroller that is used is an esp32 microcontroller which means it has Wi-Fi and

Bluetooth modules, allowing it to communicate wirelessly. The microcontroller supplies voltage (3.3 V or 5 V) and GND to three sensors: Gravity I2C O2 sensor, STC31 CO2 sensor and SM9233 differential pressure sensor.

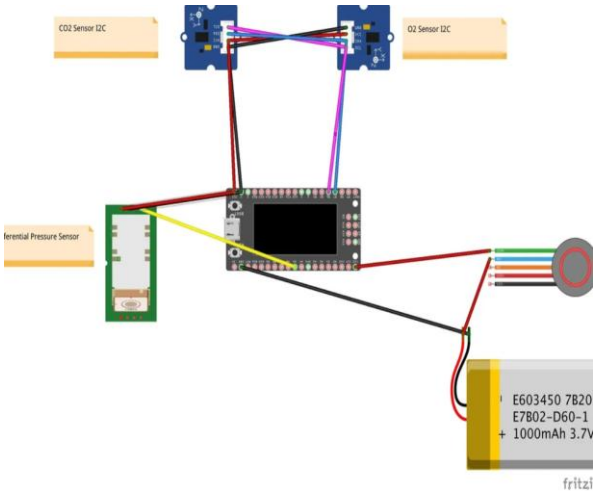


Fig1. Schematic showing the wiring of the components

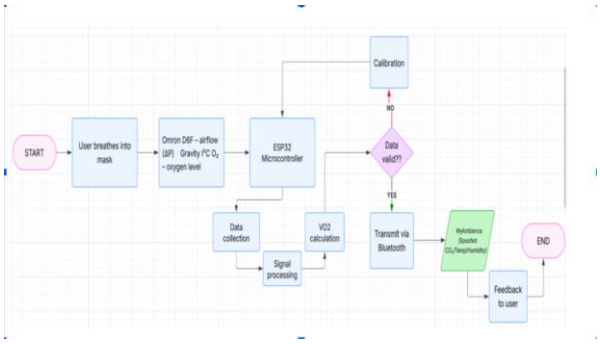


Fig2. Intended workflow for data collection and processing.

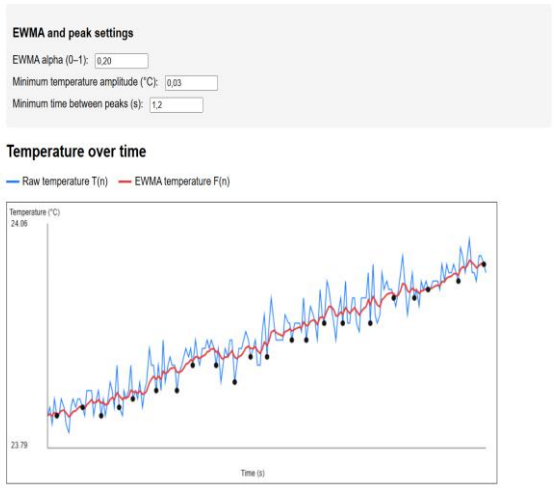


Fig3. How the respiratory rate is calculated using EWMA.

The O2 sensor is an electrochemical sensor, and it produces a voltage proportional to the concentration of O2 when coming into contact with it. The CO2 sensor is, unlike the other two, situated at the top of the venturi-tube and is a thermal conductive sensor. By measuring the temperature difference that occurs when CO2 comes into contact with the sensors it is able to determine the concentration of it. The sensor also has temperature- and RH- (relative humidity) functions, which means it can measure these two parameters as well. The differential pressure sensor measures the pressure difference between two points and is connected to the tube through two holes.

From these readings (i.e. CO_2-concentration, temperature, relative humidity, O_2-concentration and differential pressure) we derive other parameters necessary in order to calculate VO_2 MAX and other values [5]. From the temperature readings we are able to determine the respiratory frequency, from the CO_2-readings the metabolic load and from the differential pressure-readings the air volume. The formula used to calculate VO_2 MAX is:

$\rightarrow VO_2 = V_{EMean} \times (rhoBTPS/rhoSTPD) \times CO_2 \times 10.$

The V_{EMean} is the total volume of expired air in one minute, the factors rhoBTPS and rhoSTPD are used to correct for the temperature and humidity variables in our breath to get a correct measurements and the CO_2 stands for CO_2-concentration which is estimated by oxygen level loss (the actual measured CO_2-concentration is used to calculate VCO_2.

Accompanied the prototype is the Sensirion Myambience App with which the user can follow their performance values (e.g. CO_2-concentration) in real-time. This is done by utilising the Bluetooth function in the ESP32 MCU used in the project.

4. Evaluation

The evaluation focused on sensor communication and data acquisition in the inherited VO_2max mask prototype. We used the Arduino IDE Serial Monitor to inspect sensor initialisation and measurement output. After replacing wires and improving electrical connections, the sensors were detected and readings appeared. Initially negative CO_2 readings later became positive after troubleshooting, but their accuracy could not be verified against reference measurements.

Results: After replacing wires and improving the electrical connections, the sensors were detected and readings appeared. CO_2 readings were initially negative but later became positive after troubleshooting. However, their accuracy could not be verified against reference measurements. During the final troubleshooting session, CO_2 acquisition failed. The firmware displayed "CO2 sensor FAILED" and repeated "CO2 reading failed" messages, although the Bluetooth interface reported that it was ready. An I^2C diagnostic test returned error code 5, indicating a communication timeout [6]. Another scan remained at "Scanning I2C..." without completing. The cause of the communication problem could not be identified.

Limitations: We did not receive a calibration syringe, so airflow and volume measurements could not be checked against a known reference volume. The CO_2 sensor, pressure sensor, O-ring, and ESP32 T-Display required for the new assembly were also unavailable by the submission deadline. Although the 3D-printed parts were complete, the new prototype could not be fully assembled and tested.

Overall, measurement accuracy and repeatability could not be established. No validated VO_2max result or comparison with a reference metabolic system was obtained. Further evaluation requires reliable sensor communication, a completed assembly, calibration, and repeated measurements under documented conditions.

5. Conclusions

The project evaluated the existing VO_2max mask prototype and identified persistent sensor communication issues that prevented the completion of the new assembly and planned testing. Although initial sensor measurements were obtained, the prototype's measurement accuracy and reliability could not be validated, highlighting the need for further troubleshooting, calibration, and testing.

6. References

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